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Course Title:	INFO-53883 - AI/ML Capstone Project	
Evaluation Name:	INFO-53883 - D2 (15%, Group Work, Project Design Doc.)	
Team Number:	INFO-53883 (AI/ML Capstone) - Team 3B	
Referencing and Citation Style:	*APA Style* ○ *IEEE Style* ○ *N/A* ◉	
Group Members:	*First Name and Family Name of group members:*	*Signature of group members:*
	(1) Binil john (team leader)	Binil
	(2) Anisha chaudhary	Anisha
	(3) Karishma shaik	Karishma
	(4) Harmanjith singh	Harmanjith
	(5)	
Submission Date:		

AI/ML Capstone Project

D1 – Project Proposal (5% of final mark)

Due: Week 3 (softcopy on SLATE, hardcopy to your professor in class)

Name 1: _____Binil John_____ Student Number 1: _____991844827_____

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Name 5: _____ Student Number 5: _____

Instructions:

- Please work on the following deliverable with your group.
- Prepare your deliverable report as discussed in class.
- Type your report using word processing software (preferably using LaTeX, otherwise MS Word is fine).
- Upload your report in PDF format on SLATE. Team leader should upload this deliverable on SLATE as a single PDF file.
- Present a printout (stappled) of your deliverable to your professor in class in **Week 3**.

Q1: Which sector driving modern economies are you considering for your capstone project (i.e., **agriculture**, **education**, **energy**, **entertainment**, **finance**, **government**, **heath care**, **insurance**, **manufacturing**, **NGO**). Select the sector that is mostly relevant to your capstone. Briefly explain your sector selection and what experience you and your group have in this area. Answer this question in one paragraph. [**0.5 mark**]

For our capstone project, we are considering the transportation and automotive sector. Our project focuses on developing an AI-based traffic sign detection and driver alert system that can detect traffic signs in real time and provide audio alerts to improve driver awareness and road safety. We selected this sector because we have experience with python programming, machine learning, data analysis coursework and alsoAI and computer vision are increasingly being used in modern transportation and driverassistance technologies.

Q2: Identify a **real-world problem** that you and your group are interested to explore. Explain the identified "problem" in as much detail that you can. Provide citable data or context to prove that this problem exists and that it matters. Do not forget the famous saying that: "*a problem well stated is a problem half-solved*". Answer this question in two to three paragraphs. [**1 mark**]

One of the real-world problems that our group is interested to explore is road safety and the delayed detection of road hazards by drivers. Every day, drivers encounter traffic signs, pedestrians, cyclists, vehicles, construction zones, and unexpected obstacles. However, drivers can sometimes miss these hazards or notice them too late due to distractions, tiredness, poor visibility, or sudden changes on the road. When a driver does not have enough time to react, it can lead to poor driving decisions and increase the risk and severity of a collision.

Our proposed solution is an AI-based traffic sign detection and driver alert system that uses a camera and computer vision to identify traffic signs in real time. The system would detect signs such as Stop, Speed Limit, School zone .. etc. using a object detection model and provide an audio alert to the driver. This could help bring the driver's attention to important signs without requiring the driver to look at another screen. The problem is therefore not simply that traffic signs exist, but that drivers may fail to notice or respond to them in time, creating an opportunity for AI-based driver-assistance technology to provide an additional layer of awareness.

Research has shown that missing or failing to respond to traffic signs can contribute to road accidents. One IEEE/arXiv study reported that traffic-sign infractions were associated with about 12.6% of accidents caused by driver error. Another study of intersection crashes found that nearly 22% of urban crashes involved drivers running stop signs, red lights, or other traffic controls, with 39% of these crashes resulting in injuries. In addition, Transport Canada reported that distraction was a contributing factor in 17.8% of fatal collisions in Canada in 2023. These findings highlight how important driver attention and timely recognition of traffic signs are for road safety.

Q3: Identify the **prospective clients** and **stakeholders** that the real-world problem that you explained above is targeted for. What are their specific requirements? How are they currently solving this challenge without your AI/ML solution? Support your ideas with reliable market research data. Answer this question in one paragraph. [**0.5 mark**]

Targeted customers and stakeholders for our AI-based traffic sign detection and driver alert system includes individual drivers, commercial fleet owners, and transportation companies that want to improve driver awareness and reduce missed traffic signs. Insurance companies may also be interested in technologies that can support safer driving, while automakers and automotive manufacturers could explore the system as a cost-effective driver-assistance feature that can be implemented in vehicles. Government transportation agencies and road safety organizations are important partners, as the system aims to help drivers identify and react to traffic signs they might miss supports more broad public road-safety aims. These partners need a system that are capable of recognizing accurately and distinguish traffic signs from a car's live camera feed and provides the driver an audio alert instantly, and function fairly well in difficult circumstances. This issue is mainly addressed through the driver's own awareness, while some vehicles use existing ADAS traffic-sign-recognition features that operates through built-in cameras. Although commercial ADAS already use computer vision models for this purpose, our project aims to bridge the gap by attaining high and accurate detection under challenging settings in real time within minimal hardware requirements.

Q4: Explicitly justify **why an AI/ML solution is needed** here rather than applying standard heuristics or traditional software engineering alternatives? What are the benefits of using AI/ML to tackle this real-world problem? Answer this question in one paragraph. [**0.5 mark**]

A fixed rule-based approach may not work in practice since traffic signs vary in appearance with different lighting conditions, various weather conditions, viewing angle, distance, sign damage, and obstructions. In order to learn the different visual patterns that classify different sign classes, computer vison models can be trained on a large number of labelled traffic-sign images. This allows the models to generalise to signs that they have never seen before, such as rotated, fading, dimly lit, or partially visible signs. Since our system needs to work with a live video feed rather than a single image, an object detection model such as YOLO is a suitable option because it can detect and locate traffic signs in real time. This system can provide drivers with an additional layer of awareness while they are on road. Sometimes drivers may miss a sign because of lack of focus, fatigue, unfamiliar roads, or simply because the sign is difficult to notice. Our system acts as an extra set of eyes by continuously monitoring the video feed and providing an audio alert when a traffic sign is detected. AI and machine learning makes this possible as a traditional program would require an very large and continuously updated set of manually coded rules to mimic what a trained model automatically learns from the data.

Q5: Explain **how AI/ML can be integrated** into the real-world problem you identified above. Support your ideas from existing literature and/or industry benchmarks. Answer this question in two to three paragraphs. [**1 mark**]

AI and ML can be integrated into this problem With the help of computer vision based object detection that can detect and recognize traffic signs from the camera feed in real time. Instead of depending on a fixed set of hand coded rules to detect or recognize a sign. This can give drivers an additional layer of awareness, particularly when they are distracted, tired, driving on unfamiliar road or may not notice a sign. Machine learning models such as R-CNN, SSD, and YOLO have been widely used for object detection in real time. For example, one embedded ADAS study using Faster R-CNN on a Raspberry Pi achieved around 90% detection accuracy for traffic signs. This shows that traffic-sign detection can be implemented even on relatively low-cost hardware, making it possible to built an accessible driver alert system that can run on a basic hardware without requiring expensive hardware.
The main challenge in our project is to accurately detect the traffic signs in real driving conditions. Our existing system emphasizes identifying traffic signs and delivering an instant audio notification to the driver. This feels possible by referring the existing literature survey of Marques et al. demonstrating a YOLOv3 based system to detect and recognize both traffic sign and traffic lights for autonomous vehicles, it tested across simulations and real world environments under different weather conditions. On hardware side Cleofas-Sánchez et al. shows faster R-CNN deployed on a raspberry pi achieves around 90% accuracy. This is a relevant evidence that real time traffic sign detection is possible without expensive hardware. Handling challenging conditions like nighttime driving, rain, fog, glare and partially blocked signs will be important area for future work.

Q6: The AI/ML capstone project expects groups to work with **datasets**. In general, groups can **train, validate, and test an AI/ML model** of their choice. Alternatively, groups may consider **customizing an SLM or LLM with specific or proprietary data** using techniques like fine-tuning or RAG. Provide some insights on where will your data come from? Is the data available in open-source, do you need to scrape, generate it on your own, or synthetically simulate it? Provide explanation on licensing, data privacy, and the volume sufficiency needed for training, model adaptation or context injection. Also, is the considered dataset properly labelled or not and what implications will this have on your overall work. Answer this question in three to four paragraphs. [**1 mark**]

Dataset is one of the most important things to get right before we start building the model, because we need signs specific to certain classes particularly Ontario/Canadian traffic signs that are not fully available as ready-made, labeled sets on Kaggle or other public dataset platforms. Datasets like GTSRB and Mapillary Traffic Sign Dataset are built around German and global sign standards respectively, and while they're useful as a starting point or for pretraining, they don't reliably cover the specific sign designs, wording, and layouts used on Ontario roads. Because of this, we will need to build our own dataset rather than relying entirely on an existing one.

Our approach will combine two sources: manually capturing images of real traffic signs around Ontario ourselves, and supplementing this with relevant images pulled from different existing public datasets where the sign types overlap or are close enough to be useful. Where we do pull from external datasets like GTSRB or Mapillary, For licensing GTSRB allows academic use with a citation, and Mapillary has its own research-use terms. Since this project is for a class, this works fine. Images we capture ourselves don't have this issue, since it's our own data.

Once we've gathered the raw data, we need to annotate it drawing bounding boxes around each sign in every image and assigning it the correct class label since YOLO requires labeled data in this format to train on. A small batch of images is unlikely to be enough for the model to learn traffic signs reliably. We expect to need at least a few hundred labeled images for each sign class, with additional examples for signs that have more visual variation, such as different speed-limit numbers or signs that look similar to one another. Having enough examples can help reduce overfitting and improve the model's ability to recognize signs it has not seen before. To increase the size and variety of our dataset without having to manually collect every possible example, we plan to use data augmentation techniques such as rotation, brightness and contrast adjustments, blurring, cropping, and adding noise. More challenging conditions such as rain, fog, and night-time driving can also be considered as part of future improvements to the system.

On privacy: since we'll be capturing our own footage/images around Ontario, we are not going to store any of the captured video footage. The camera footage will only be used temporarily for real-time detection. The video will pass through the model to identify traffic signs and generate an audio alert, but it will not be saved, logged, or stored afterward. This helps reduce privacy concerns related to accidentally capturing information such as faces, license plates, or pedestrians, since the live footage does not remain in the system after detection. For images that are specifically collected and used to train the model, we will review the dataset before using it. Where necessary, identifiable information such as faces and license plates will be blurred or anonymized to help protect individuals' privacy.

Q7: An AI/ML technology stack is generally composed of the **infrastructure layer** (i.e., HW/SW), **data layer** (i.e., injection/cleaning/storage), **framework and model layer** (i.e., TensorFlow/PyTorch, LLM/SLM, fine-tuning/RAG), **MLOps layer** (e.g., Hugging Face, Docker), and the **application layer** (e.g., APIs, UI/Streamlit). Identify a *preliminary* technology stack suited to tackle the AI/ML challenge of your capstone project. Answer this question by providing a brief sentence for each AI/ML technology stack. [**0.5 mark**]

Infrastructure layer (HW/SW): We will use our GPU-enabled laptop or virtual GPU (google colab) for model training, and a standard laptop/webcam setup with Python and OpenCV for real-time inference during testing and demonstration.
Data layer (ingestion/cleaning/storage):. We will use traffic-sign images captured in Ontario, along with additional images from publicly available datasets for ingestion, manually annotated with bounding boxes/class labels (e.g., using LabelImg or Roboflow), cleaned to remove duplicates or poor-quality images, and stored in a structured folder/CSV format compatible with YOLO's training pipeline, with augmentation applied to expand dataset volume.
Framework and model layer: We'll use PyTorch as the underlying deep learning framework, with YOLO and other algorithms as the pretrained object detection model, which we'll fine-tune on our custom traffic-sign dataset rather than training from scratch.
MLOps layer: We'll use Roboflow or Hugging Face for dataset versioning and management, and a Python virtual environment to keep training and inference dependencies consistent and reproducible across team members' machines.
Application layer: We'll build the user-facing interface with Streamlit, integrating OpenCV for webcam video capture and frame processing, and a text-to-speech library either pyttsx3 or gTTS to generate the real-time audio alerts when a traffic sign is detected.

References (Using IEEE style, include 5-7 peer-reviewed sources to back your AI/ML capstone

[1] A. M. Jenifer and R. Balamanigandan, "Advanced Road Sign Detection Techniques: Leveraging Harris Corner Detector for Robust Recognition," 2024 4th International Conference on Soft Computing for Security Applications (ICSCSA), Salem, India, 2024, pp. 345-350, doi: 10.1109/ICSCSA64454.2024.00061.

[2 Y. Zhu, "Road Traffic Sign Detection Algorithm Based on Improved YOLOv8n," 2026 7th International Seminar on Artificial Intelligence, Networking and Information Technology (AINIT), Dalian, China, 2026, pp. 755-758, doi: 10.1109/AINIT70033.2026.11557703.

[3] Y. Setiawan, M. N. Puji and W. Astuti, "Traffic Signs Detection System Using YOLO (You Only Look Once) That Provides Notification," 2024 IEEE International Conference on Smart Mechatronics (ICSMech), Yogyakarta, Indonesia, 2024, pp. 95-100, doi: 10.1109/ICSMech62936.2024.10812285.

[4] L. Qiu, "Real-Time Traffic Sign Detection System for Autonomous Driving Based on YOLO Algorithm," 2024 Cross Strait Radio Science and Wireless Technology Conference (CSRSWTC), Macao, China, 2024, pp. 1-4, doi: 10.1109/CSRSWTC64338.2024.10811588.

[5] Tian and J. Xiong, "A Lightweight Real-Time Traffic Sign Detection Model Based on Improved YOLOv8n," 202544th Chinese Control Conference (CCC), Chongqing, China, 2025, pp. 8501-8506, doi: 10.23919/CCC64809.2025.11178591.

Good Luck!